

Revised Generation Interconnection System Impact Study Report

For Queue Project Q22-2267

Shullsburg 138 kV
March 23, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2267. The Interconnection Customer proposes a 194.9 MW solar photovoltaic facility interconnecting at the Shullsburg 138 kV substation in the Virginia area. The request was received on December 28, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 663 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,610,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: HARTMAN CRK SUBSTATION 345-1 kV XFMR
Contingency: LOWMOOR – HINTON AP 345 KV
Loading: 107.7% (Rating: 630 MVA)
DFAX: 44.1%
Pre-project loading: 94.0%
2. Facility: HARTMAN CREEK 138/1 kV Transformer
Contingency: LOWMOOR – HINTON AP 345 KV L
Loading: 107.7% (Rating: 630 MVA)
DFAX: 44.1%
Pre-project loading: 94.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HARTMAN CRK 345-1 kV Autotransformer	Replace transformer with higher-rated unit	\$11,500,000	10% (\$1,113,000)
HARTMAN CRK 138/1kV Auto	Replace transformer with higher-rated unit	\$5,133,000	10% (\$497,000)
	Total allocated to project		\$1,610,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.